The Simple Regression Model

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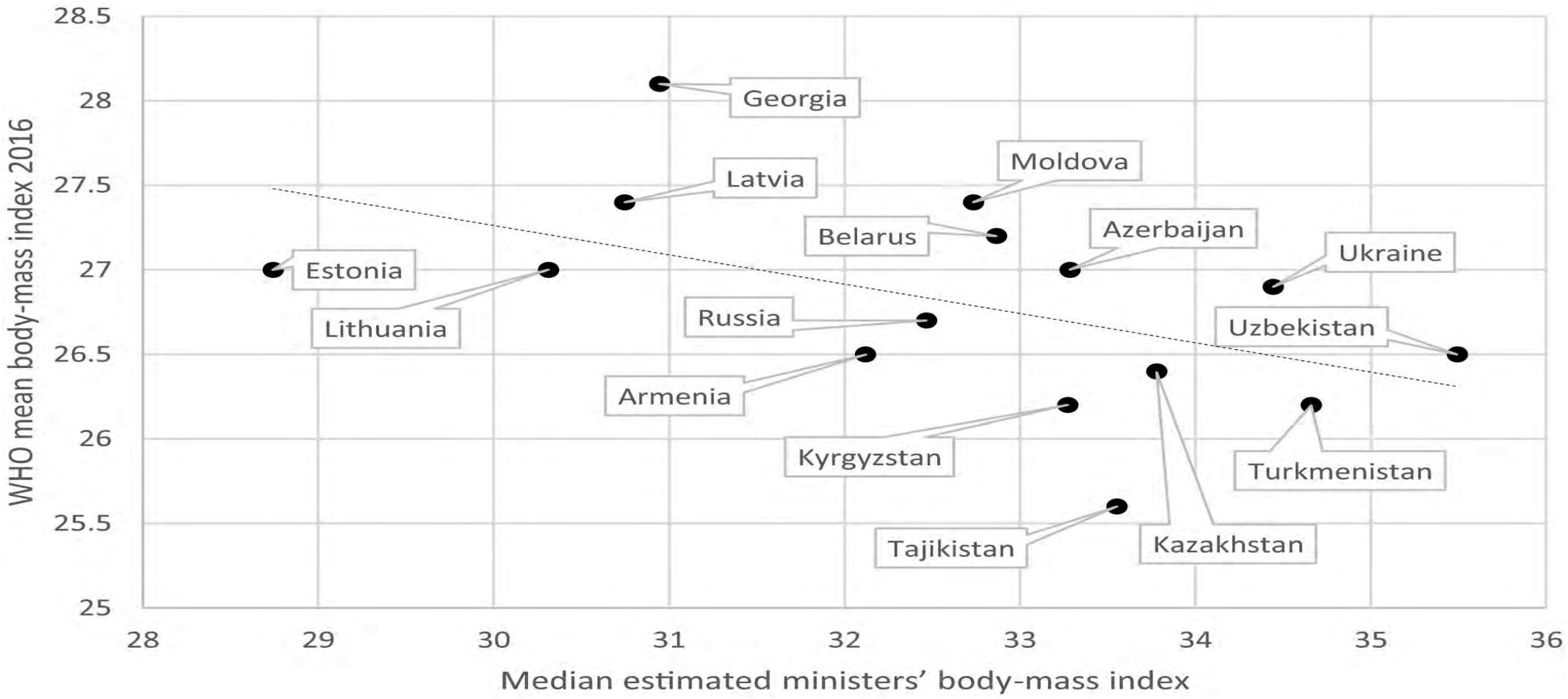
Outline

- Definition of the Simple Regression Model
- Deriving the Ordinary Least Squares Estimates
- Properties of OLS on any Sample of Data
- Units of Measurement and Functional Form
- Testing
- Confidence interval

Something fun to start with

Country	Median estimated ministers' BMI	Image of a minister with median BMI	Corruption Perceptions Index 2017	World Bank Control of Corruption 2017	Index of Public Integrity 2017	IDEA Absence of Corruption 2017
Estonia	28.7		71	1.24	8.93	0.807
Lithuania	30.3		59	0.55	7.82	0.641
Latvia	30.7		58	0.54	7.93	0.631
Georgia	30.9		56	0.74	7.30	0.652
Armenia	32.1		35	−0.56		0.352
Russia	32.5		29	−0.89	5.90	0.249
Moldova	32.7		31	−0.80	6.44	0.370

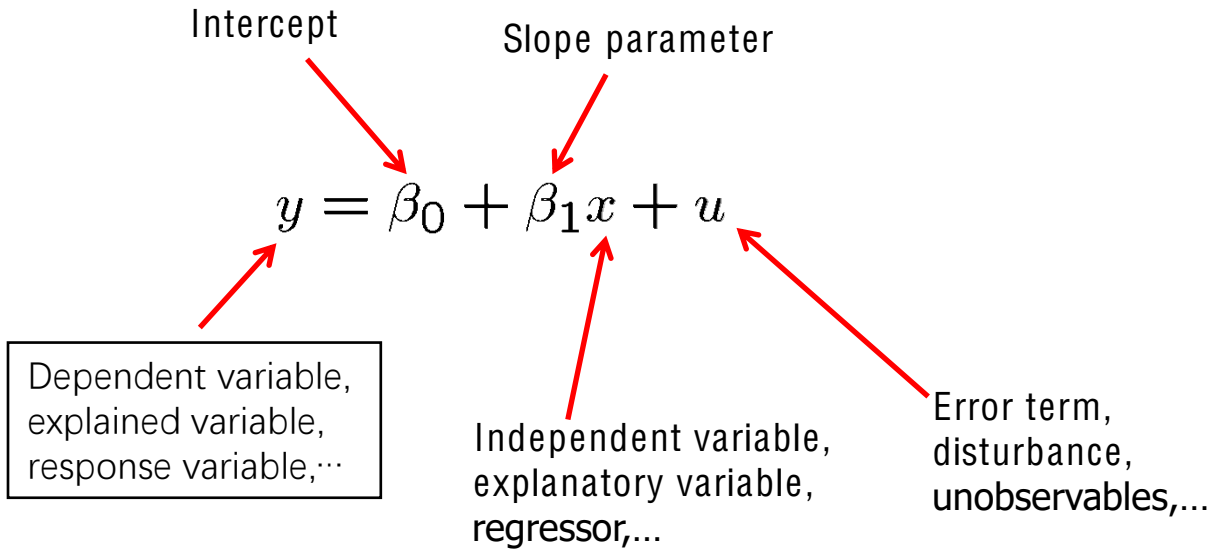
Obesity of politicians and corruption



Definition of the simple linear regression model

- Definition of the simple linear regression model
- For instance, is amount of fertilizer, and is soybean yield; is years of schooling, is hourly wage.

"Explains variable y in terms of variable x"



Definition of the simple linear regression model

- Interpretation of the simple linear regression model
- All other factors that affect are in u. We want to know how y changes when x changes, holding u fixed.

"Studies how y varies with changes in x:"

$$\frac{\Delta y}{\Delta x} = \beta_1 \quad \text{as long as} \quad \frac{\Delta u}{\Delta x} = 0$$

By how much does the dependent variable change if the independent variable is increased by one unit?

Interpretation only correct if all other things remain equal when the independent variable is increased by one unit

- The simple linear regression model is rarely applicable in practice but its discussion is useful for pedagogical reasons

Definition of the simple linear regression model

- Example: Soybean yield and fertilizer

$$yield = \beta_0 + \beta_1 fertilizer + u$$

Rainfall,
land quality,
presence of parasites, ...

Measures the effect of fertilizer on yield, holding all other factors fixed

- Example: A simple wage equation

$$wage = \beta_0 + \beta_1 educ + u$$

Labor force experience,
tenure with current employer,
work ethic, intelligence, ...

Measures the change in hourly wage given another year of education, holding all other factors fixed

When is there a causal interpretation?

- **When is there a causal interpretation?**

➢How can we hope to learn in general about the ceteris paribus effect of x on y, holding other factors fixed, when we are ignoring all those other factors?

➢What if x and u are related?

- **Example: wage equation**

$$wage = \beta_0 + \beta_1 educ + u$$ ← e.g. intelligence …

The conditional mean independence assumption is unlikely to hold because individuals with more education will also be more intelligent on average.

Expected value of the OLS estimators

- **The four assumptions:**

SLR.1: $y = \beta_0 + \beta_1 x + u$

SLR.2: random sampling from the population

SLR.3: some sample variation in the x_i

SLR.4: $E(u|x) = 0$ → $cov(x, u) = 0$ ②

$E(u) = E[E(u|x)] = 0$

$E(u) = 0$ ①

$E[x - E(x)] \cdot [u - E(u)] = E[(x - E(x))u] = E(xu) - E[E(x) \cdot u]$

$= E(xu) - E[E(xu|x)] = E[x \cdot E(u|x)] = 0$

the law of iterated expectation

Expected value of the OLS estimators

- **Standard assumptions for the linear regression model**
- **Assumption SLR.1 (Linear in parameters)**

$$y = \beta_0 + \beta_1 x + u$$

In the population, the relationship between y and x is linear

- **Assumption SLR.2 (Random sampling)**

$$\{(x_i, y_i) : \ i = 1, \ldots n\}$$

The data is a random sample drawn from the population

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

Each data point therefore follows the population equation

Expected value of the OLS estimators

- **Discussion of random sampling: Wage and education（Appendix C）**
  - The population consists, for example, of all workers of Shenzhen
  - In the population, a linear relationship between wages (or log wages) and years of education holds
  - Draw completely randomly a worker from the population
  - The wage and the years of education of the worker drawn are random because one does not know beforehand which worker is drawn
  - Repeat random draw n times
  - The wages and years of education of the sampled workers are used to estimate the linear relationship between wages and education

Expected value of the OLS estimators

- **Assumptions for the linear regression model (cont.)**
- **Assumption SLR.3 (Sample variation in explanatory variable)**

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 > 0$$

The values of the explanatory variables are not all the same (otherwise it would be impossible to study how different values of the explanatory variable lead to different values of the dependent variable)

- **Assumption SLR.4 (Zero conditional mean)**

$$E(u_i|x_i) = 0$$

The value of the explanatory variable must contain no information about the mean of the unobserved factors

This is the key assumption for showing that OLS is **unbiased**

When is there a causal interpretation?

- **Zero conditional mean independence assumption**

  1. Normalizing u

$$E(u) = 0$$

  2. u is mean independent of x

$$E(u|x) = E(u)$$

  if the unobservable is the same across all slices of the population determined by the value of x .

  Combined:

$$E(u|x) = 0$$

  ← The explanatory variable must not contain information about the mean of the unobserved factors

- **Example: education and "ability"**

Population regression function (PFR)

- Population regression function (PFR)
  - The conditional mean independence assumption implies that

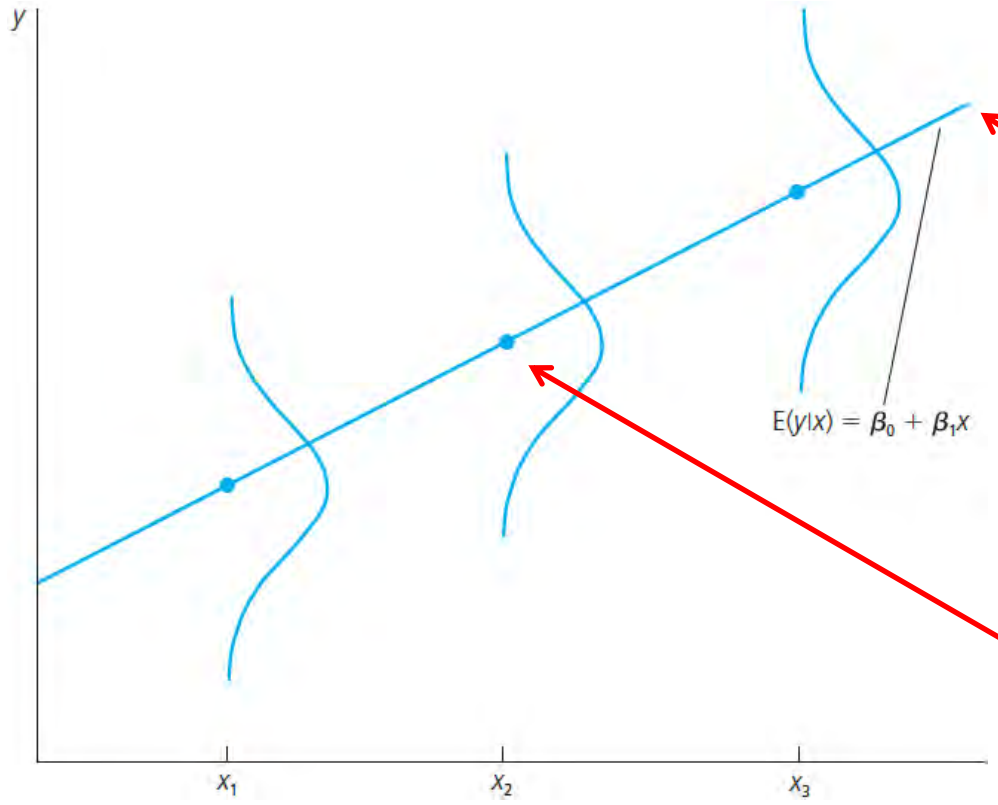
$$E(y|x) = E(\beta_0 + \beta_1 x + u|x)$$

$$= \beta_0 + \beta_1 x + E(u|x)$$

$$= \beta_0 + \beta_1 x$$

  - This means that the average value of the dependent variable can be expressed as a linear function of the explanatory variable

Population regression function (PFR)



Population regression function

For individuals with $x = x_2$, the average value of y is $\beta_0 + \beta_1 x_2$

Definition of the simple linear regression model

- **Example: College versus High School GPA.**

  Suppose for the population of students attending an university, we (somehow) know

$$E(colGPA|hsGPA) = 1.5 + 0.5\ hsGPA,$$

  If $hsGPA = 3.6$ then the **average** of $colGPA$ among students with this particular high school GPA is

$$1.5 + 0.5(3.6) = 3.3$$

- Regression analysis is essentially about explaining effects of explanatory variables on **average** outcomes of y .

Deriving the Ordinary Least Squares Estimate

- **Deriving the Ordinary Least Squares Estimate**

  In order to estimate the regression model one needs data

- A random sample of n observations

(x_1, y_1) ← First observation

(x_2, y_2) ← Second observation

⋮

(x_n, y_n) ← n-th observation

$\{(x_i, y_i) : \ i = 1, \ldots n\}$

Value of the explanatory variable of the i-th observation

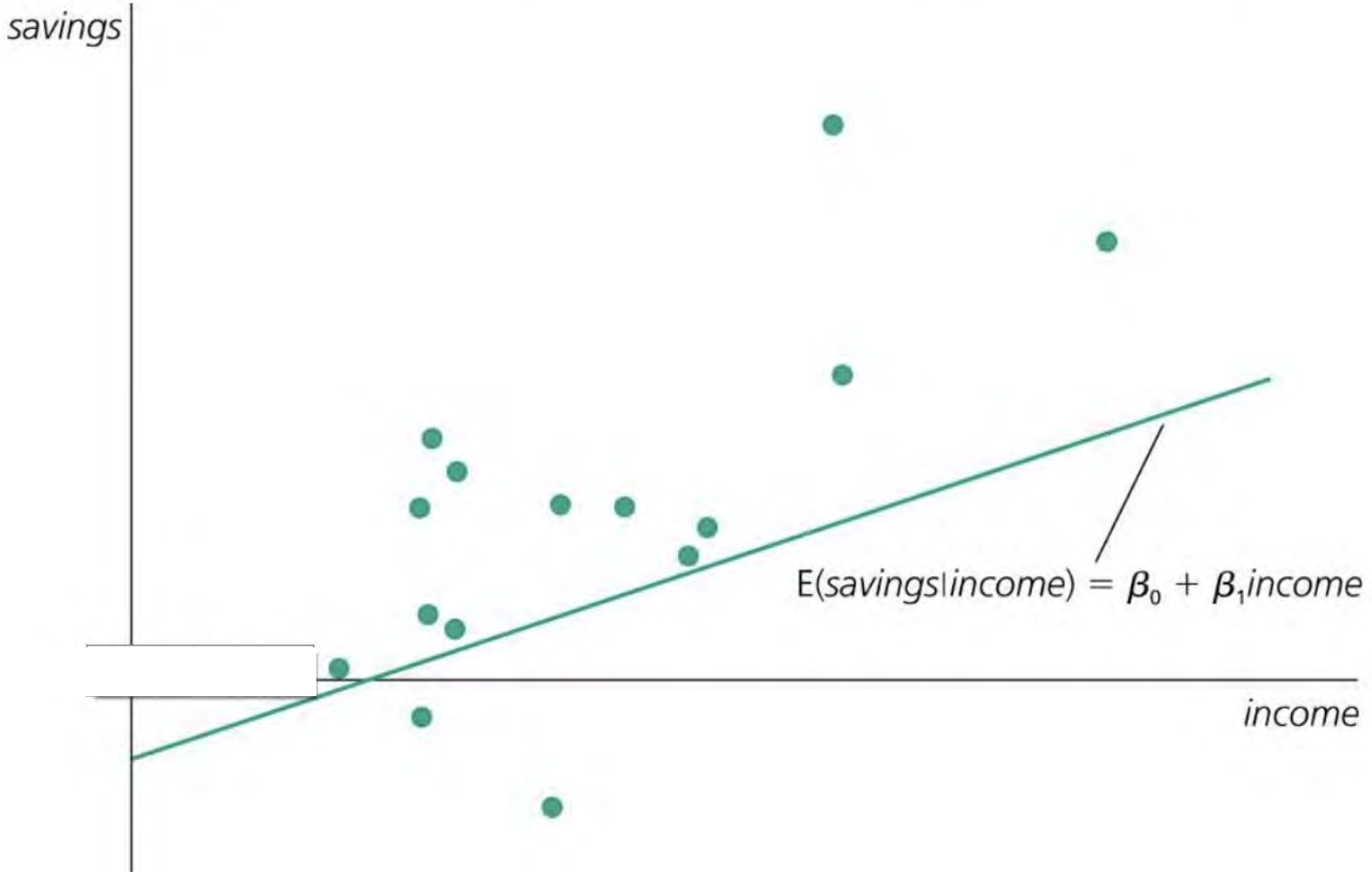
Value of the dependent variable of the i-th observation

- Using a random sample, we can generate estmators $\hat{\beta}_0$ and $\hat{\beta}_1$ to estimate the population parameters β_0 and β_1.
- We put a "hat" on the beta to denote an estimator

Deriving the Ordinary Least Squares Estimate

- **Example: Saving and income**

  This graph shows n = 15 families and the population regression of saving on income.



Deriving the Ordinary Least Squares Estimate

- **Deriving the OLS estimate**
- Where does the name "**Ordinary Least Squares**" come from?

  For any candidates $\hat{\beta}_0$ and $\hat{\beta}_1$, define a **fitted (or predicted) value** for each data point i as:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

  We have n of these. It is the value we predict for y_i given that x has taken on the value x_i.

- The mistake we make is the residual:

$$\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i,$$

  and we have n residuals.

Deriving the Ordinary Least Squares Estimate

- **What does "as good as possible" mean?**
- Regression residuals

$$\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

- Choose $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize sum of squared regression residuals

$$\min \sum_{i=1}^{n} \hat{u}_i^2 \quad \rightarrow \min \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

- First order conditions $\sum \hat{u}_i = 0$

$$n^{-1} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$n^{-1} \sum_{i=1}^{n} x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Deriving the Ordinary Least Squares Estimate

- **First equation**

$$
\begin{aligned}
n^{-1}\sum_{i=1}^{n}(y_i-\hat{\beta}_0-\hat{\beta}_1x_i) &= n^{-1}\sum_{i=1}^{n}y_i-n^{-1}\sum_{i=1}^{n}\hat{\beta}_0-n^{-1}\sum_{i=1}^{n}\hat{\beta}_1x_i \\
&= n^{-1}\sum_{i=1}^{n}y_i-\hat{\beta}_0-\hat{\beta}_1\left(n^{-1}\sum_{i=1}^{n}x_i\right) \\
&= \bar{y}-\hat{\beta}_0-\hat{\beta}_1\bar{x}=0
\end{aligned}
$$

implies

$$\bar{y}=\hat{\beta}_0+\hat{\beta}_1\bar{x}$$

and plug this into the **second equation**

$$\sum_{i=1}^{n}x_i[y_i-(\bar{y}-\hat{\beta}_1\bar{x})-\hat{\beta}_1x_i]=0$$

Deriving the Ordinary Least Squares Estimate

Simple algebra gives

$$\sum_{i=1}^{n} x_i(y_i - \bar{y}) = \hat{\beta}_1 \left[\sum_{i=1}^{n} x_i(x_i - \bar{x}) \right] \qquad (1)$$

Because

$$\sum_{i=1}^{n} (x_i - \bar{x}) = 0$$

$$\sum_{i=1}^{n} x_i(y_i - \bar{y}) = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} (x_i - \bar{x}) y_i$$

$$\sum_{i=1}^{n} x_i(x_i - \bar{x}) = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$\frac{\sum_{i=1}^{n} (X_i - \bar{x}) Y_i}{\sum_{i=1}^{n} X_i (X_i - \bar{x})}$

$\Sigma (X_i - \bar{x})(Y_i - \bar{y})$
$= \Sigma (X_i - \bar{x}) Y_i$
$= \Sigma X_i (Y_i - \bar{y})$

one can use equation (1) to derive:

- **Ordinary Least Squares (OLS) estimates**

$\frac{\sum_{i=1}^{n} X_i (Y_i - \bar{y})}{\sum_{i=1}^{n} X_i (X_i - \bar{x})}$ $= \hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

$= \frac{cov(x,y)}{Var(x,y)}$

Deriving the Ordinary Least Squares Estimate

- Once we have the numbers $\hat{\beta}_0$ and $\hat{\beta}_1$ for a given data set, we write the **OLS regression line** as a function of x:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

- The OLS regression line allows us to predict y for any (sensible) value of x. It is also called the **sample regression function**.

  - The intercept, $\hat{\beta}_0$, is the predicted y when $x = 0$.

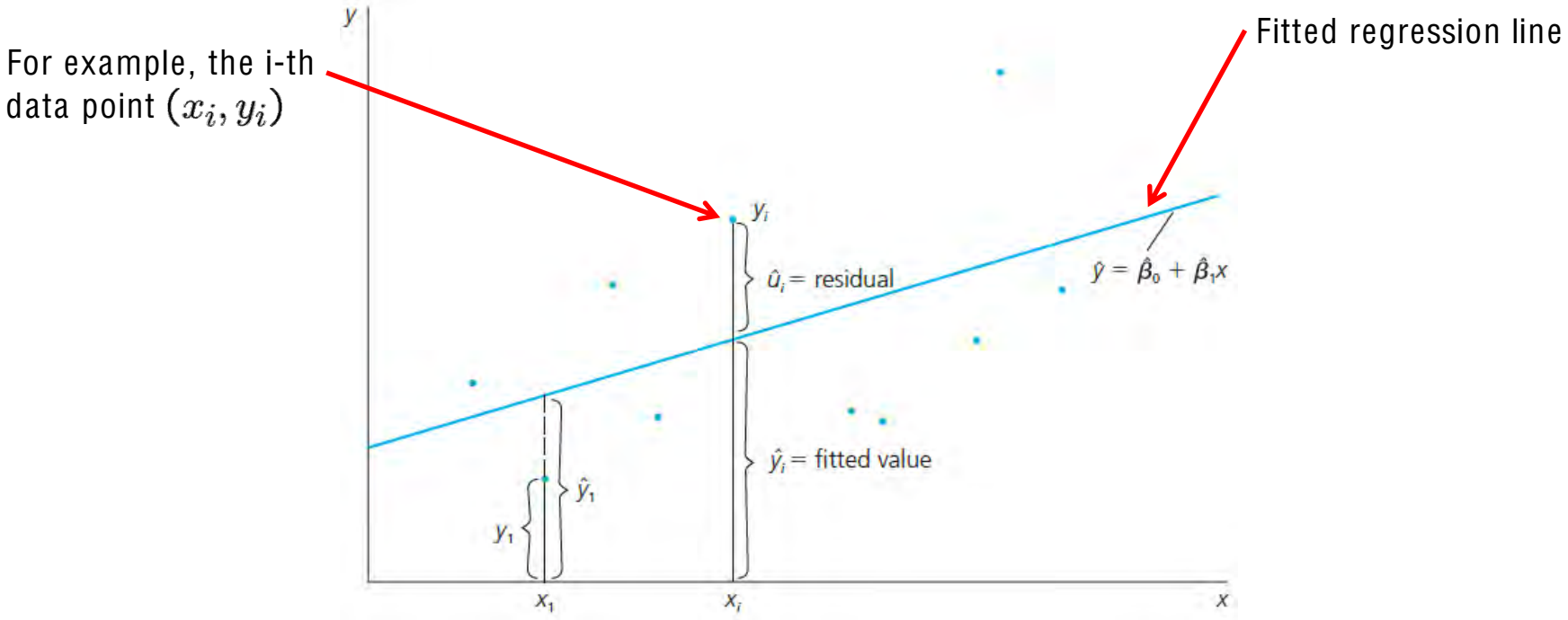
  - The slope, $\hat{\beta}_1$, allows us to predict changes in y for any (reasonable) change in x, so

$$\Delta\hat{y} = \hat{\beta}_1 \Delta x$$

If $\Delta x = 1$, so that x increases by one unit, then $\Delta\hat{y} = \hat{\beta}_1$.

Deriving the Ordinary Least Squares Estimate

- Fit as good as possible a regression line through the data points:



Deriving the Ordinary Least Squares Estimate

- CEO Salary and return on equity

$$salary = \beta_0 + \beta_1 roe + u$$

Salary in thousands of dollars

Average return on equity of the CEO's firm

- Fitted regression

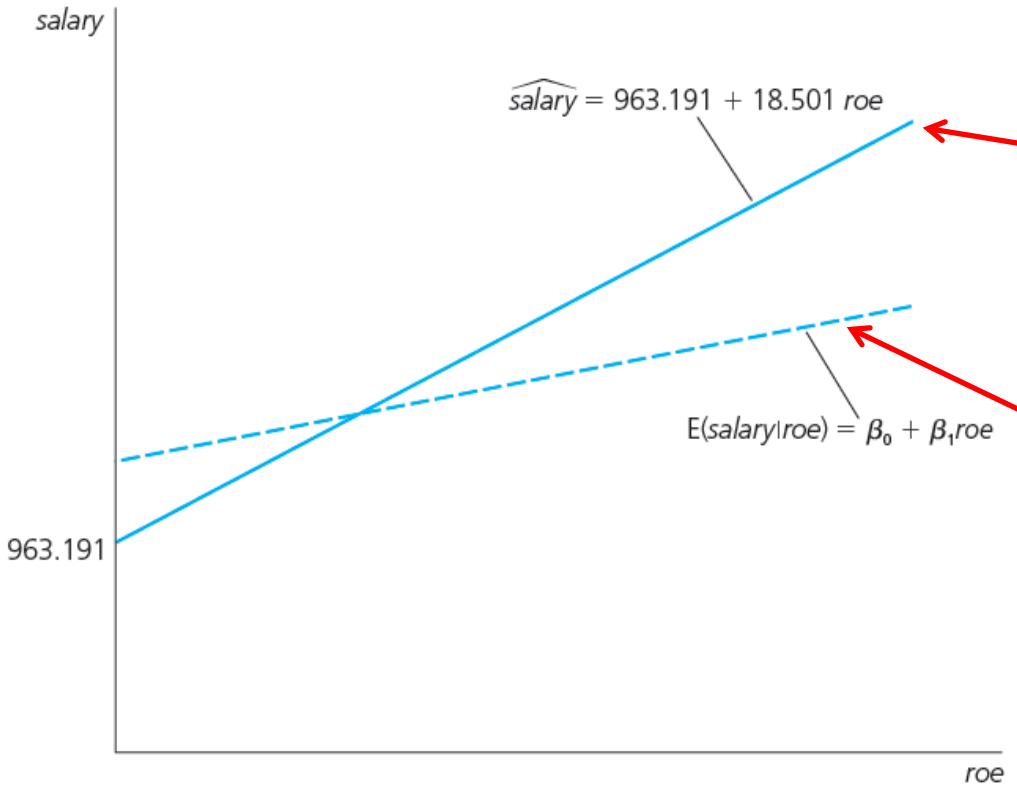
$$\widehat{salary} = 963.191 + 18.501\ roe$$

Intercept

If the return on equity increases by 1 percent, then salary is predicted to change by $18,501

- Causal interpretation?

Deriving the Ordinary Least Squares Estimate



Fitted regression line
(depends on sample)

Unknown population regression line

Deriving the Ordinary Least Squares Estimate

- Wage and education

$$wage = \beta_0 + \beta_1 educ + u$$

Hourly wage in dollars

Years of education

- Fitted regression

$$\widehat{wage} = -0.90 + 0.54\ educ$$

Intercept

In the sample, one more year of education was associated with an increase in hourly wage by $0.54

- Causal interpretation?

Deriving the Ordinary Least Squares Estimate

- Voting outcomes and campaign expenditures (two parties)

$$voteA = \beta_0 + \beta_1 shareA + u$$

Percentage of vote for candidate A

Percentage of campaign expenditures candidate A

- Fitted regression

$$\widehat{voteA} = 26.81 + 0.464\ shareA$$

Intercept

If candidate A's share of spending increases by one percentage point, he or she receives 0.464 percentage points more of the total vote

- Causal interpretation?

Properties of OLS on any sample of data

- Properties of OLS on any sample of data
- Fitted values and residuals

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \qquad \hat{u}_i = y_i - \hat{y}_i$$

Fitted or predicted values

Deviations from regression line (= residuals)

- Algebraic properties of OLS regression

$$\sum_{i=1}^{n} \hat{u}_i = 0$$

Deviations from regression line sum up to zero

$$\sum_{i=1}^{n} x_i \hat{u}_i = 0$$

Covariance between deviations and regressors is zero

$$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

Sample averages of y and x lie on regression line

Properties of OLS on any sample of data

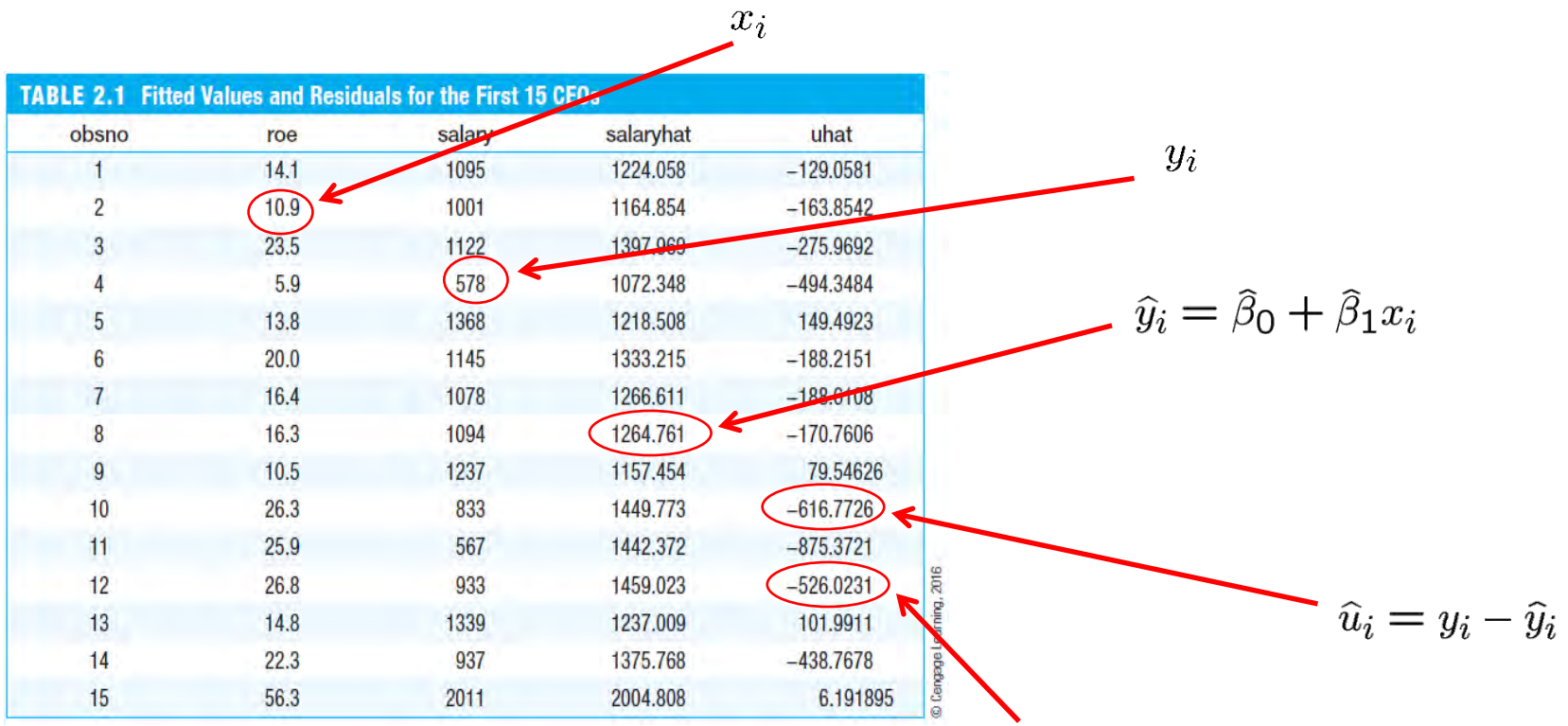


TABLE 2.1 Fitted Values and Residuals for the First 15 CEOs

obsno	roe	salary	salaryhat	uhat
1	14.1	1095	1224.058	−129.0581
2	10.9	1001	1164.854	−163.8542
3	23.5	1122	1397.969	−275.9692
4	5.9	578	1072.348	−494.3484
5	13.8	1368	1218.508	149.4923
6	20.0	1145	1333.215	−188.2151
7	16.4	1078	1266.611	−188.6108
8	16.3	1094	1264.761	−170.7606
9	10.5	1237	1157.454	79.54626
10	26.3	833	1449.773	−616.7726
11	25.9	567	1442.372	−875.3721
12	26.8	933	1459.023	−526.0231
13	14.8	1339	1237.009	101.9911
14	22.3	937	1375.768	−438.7678
15	56.3	2011	2004.808	6.191895

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x_i

y_i

$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$

$\hat{u}_i = y_i - \hat{y}_i$

For example, CEO number 12‘s salary was $526,023 lower than predicted using the the information on his firm‘s return on equity

Expected value of the OLS estimators

- **Expected values and variances of the OLS estimators**
- **The estimated regression coefficients are random variables because they are calculated from a random sample**

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Data is random and depends on particular sample that has been drawn

- The question is what the estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ will estimate on average and how large their variability in repeated samples is.

$$E(\hat{\beta}_0) = ?, \; E(\hat{\beta}_1) = ? \qquad Var(\hat{\beta}_0) = ?, \; Var(\hat{\beta}_1) = ?$$

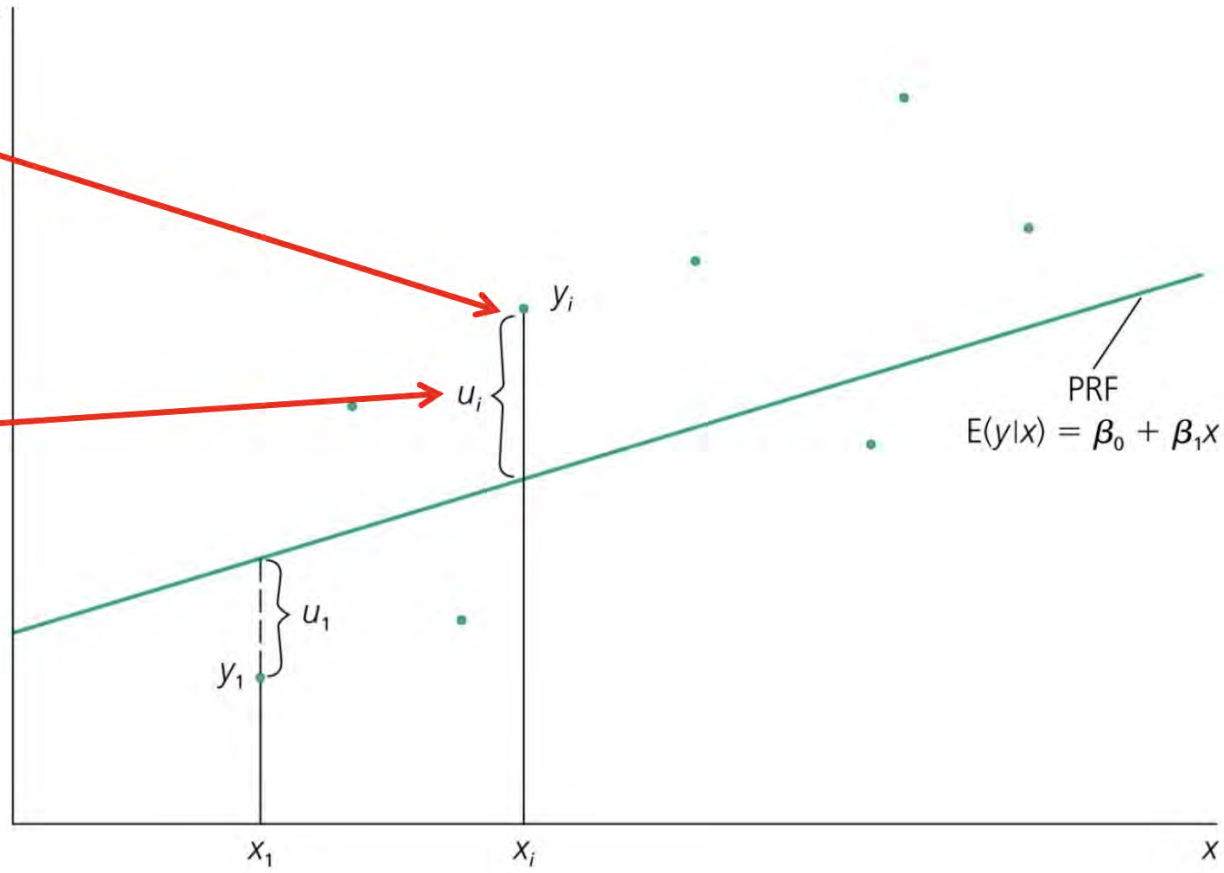
Expected value of the OLS estimators

- **Graph of** $y_i = \beta_0 + \beta_1 x_i + u_i$

The values drawn for the i-th worker (x_i, y_i)

The implied deviation from the population relationship for the i-th worker:

$u_i = y_i - \beta_0 - \beta_1 x_i$



Expected value of the OLS estimators

- **Unbiasedness of OLS**

$$E(\hat{\beta}_1) = \beta_1$$

where the expected value means averaging across random samples

- **Step 1**: Write down a formula for $\hat{\beta}_1$. It is convenient to use

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})y_i}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

It is convenient to define $SST_x = \sum_{i=1}^{n}(x_i - \bar{x})^2$, to total variation in the x_i , and write

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})y_i}{SST_x}$$

Remember, SST_x is just some positive number. The existence of $\hat{\beta}_1$ follows from SLR.3.

Expected value of the OLS estimators

- **Step 2**: Replace each y_i with $y_i = \beta_0 + \beta_1 x_i + u_i$ (which uses SLR.1 and the fact that we have data from SLR.2).

  The numerator becomes

$$\begin{aligned}
\sum_{i=1}^{n}(x_i - \bar{x})y_i &= \sum_{i=1}^{n}(x_i - \bar{x})(\beta_0 + \beta_1 x_i + u_i) \\
&= \beta_0 \sum_{i=1}^{n}(x_i - \bar{x}) + \beta_1 \sum_{i=1}^{n}(x_i - \bar{x})x_i + \sum_{i=1}^{n}(x_i - \bar{x})u_i \\
&= 0 + \beta_1 \sum_{i=1}^{n}(x_i - \bar{x})^2 + \sum_{i=1}^{n}(x_i - \bar{x})u_i \\
&= \beta_1 SST_x + \sum_{i=1}^{n}(x_i - \bar{x})u_i
\end{aligned}$$

Expected value of the OLS estimators

- We have shown

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n}(x_i - \bar{x})u_i}{SST_x}$$

Now define

$$w_i = \frac{(x_i - \bar{x})}{SST_x}$$

so we have

$$\hat{\beta}_1 = \beta_1 + \sum_{i=1}^{n} w_i u_i$$

$E(\hat{\beta}_1) = 0 + \beta_1 + E(\Sigma w_i \mu_i)$

$= \beta_1 + \Sigma E(w_i \mu_i)$ → w is a function of x

$= \beta_1 + \Sigma E[E(w_i \mu_i | x_i)]$

$= \beta_1 + \Sigma E[w_i E(\mu_i | x)] = \beta_1 + 0$

- $\hat{\beta}_1$ is a linear function of the unobserved errors, u_i. The w_i are all functions of $\{x_1, x_2, ..., x_n\}$
- The (random) difference between $\hat{\beta}_1$ and β_1 is due to this linear function of the unobservables.

Expected value of the OLS estimators

- **Step 3**: Find $E(\hat{\beta}_1)$.

  Under Assumptions SLR.2 and SLR.4, $E(u_i|x_1, x_2, ..., x_n) = 0$ .That means, conditional on $\{x_1, x_2, ..., x_n\}$ (and using SLR.3)

$$E(w_i u_i) = w_i E(u_i) = 0$$

because w_i is a function of $\{x_1, x_2, ..., x_n\}$.

$$\begin{aligned} E(\hat{\beta}_1) &= E\left(\beta_1 + \sum_{i=1}^{n} w_i u_i\right) \\ &= \beta_1 + \sum_{i=1}^{n} E(w_i u_i) = \beta_1 + \sum_{i=1}^{n} w_i E(u_i) \\ &= \beta_1 \end{aligned}$$

Since unbiasedness holds for any outcome on $\{x_1, x_2, ..., x_n\}$, unbiasedness also holds without conditioning on$\{x_1, x_2, ..., x_n\}$

Expected value of the OLS estimators

- **Theorem 2.1 (Unbiasedness of OLS)**

$E(\hat{\beta}_0) = E(\bar{y} - \hat{\beta}_1 \bar{x}) = \bar{y} - \bar{x} \cdot E(\hat{\beta}_1) = \hat{y} - \beta_1 \bar{x} = \beta_0$

$$SLR.1 - SLR.4 \quad \Rightarrow \quad E(\hat{\beta}_0) = \beta_0, \; E(\hat{\beta}_1) = \beta_1$$

- **Interpretation of unbiasedness**
  - The estimated coefficients may be smaller or larger, depending on the sample that is the result of a random draw
  - However, **on average**, they will be equal to the values that characterize the true relationship between y and x in the population
  - "On average" means if sampling was repeated, i.e. if drawing the random sample und doing the estimation was repeated many times
  - In a given sample, estimates may differ considerably from true values



Expected value of the OLS estimators

- **Simulation in Stata:**

$$y = 3 + 2x + u$$

$x \sim Normal(0, 9)$, $u \sim Normal(0, 36)$, and they are independent.

- If we repeat the experiment again and again, and average the $\hat{\beta}_1$, we would get very close to 2.
- The problem is, we do not know which kind of sample we have. We can never know whether we are close to the population value.
- Generally, we hope that our sample is "Typical" and produces a slope estimate close to β_1 but we can never know.

Variances of the OLS estimators

- **Variances of the OLS estimators**
  - Depending on the sample, the estimates will be nearer or farther away from the true population values
  - How far can we expect our estimates to be away from the true population values on average (= sampling variability)?
  - Sampling variability is measured by the estimator's variances

$$Var(\hat{\beta}_0),\ Var(\hat{\beta}_1)$$

- **Assumption SLR.5 (Homoskedasticity, or Constant Variance)**

The error has the same variance given any value of the explanatory variable x :

$$Var(u_i|x_i) = \sigma^2$$

The value of the explanatory variable must contain no information about the variability of the unobserved factors

Variances of the OLS estimators

- Because we assume SLR.4, that is, $E(u|x) = 0$, whenever we assume SLR.5, we can also write

$$E(u^2|x) = \sigma^2 = E(u^2)$$

- Under the population Assumptions SLR.1 $(y = \beta_0 + \beta_1 x + u)$, SRL.4 $(E(u|x) = 0)$ and SLR.5 $(Var(u|x) = \sigma^2)$,

$$\begin{aligned} E(y|x) &= \beta_0 + \beta_1 x \\ Var(y|x) &= \sigma^2 \end{aligned}$$

- So the average or expected value of y is allowed to change with x -- in fact, this is what interests us -- but the variance does not change with x.

Variances of the OLS estimators

- **Example: University GPA vs. High school GPA**

  If we impose SLR.1 — SLR.4

$$E(colGPA|hsGPA) = \beta_0 + \beta_1 hsGPA$$

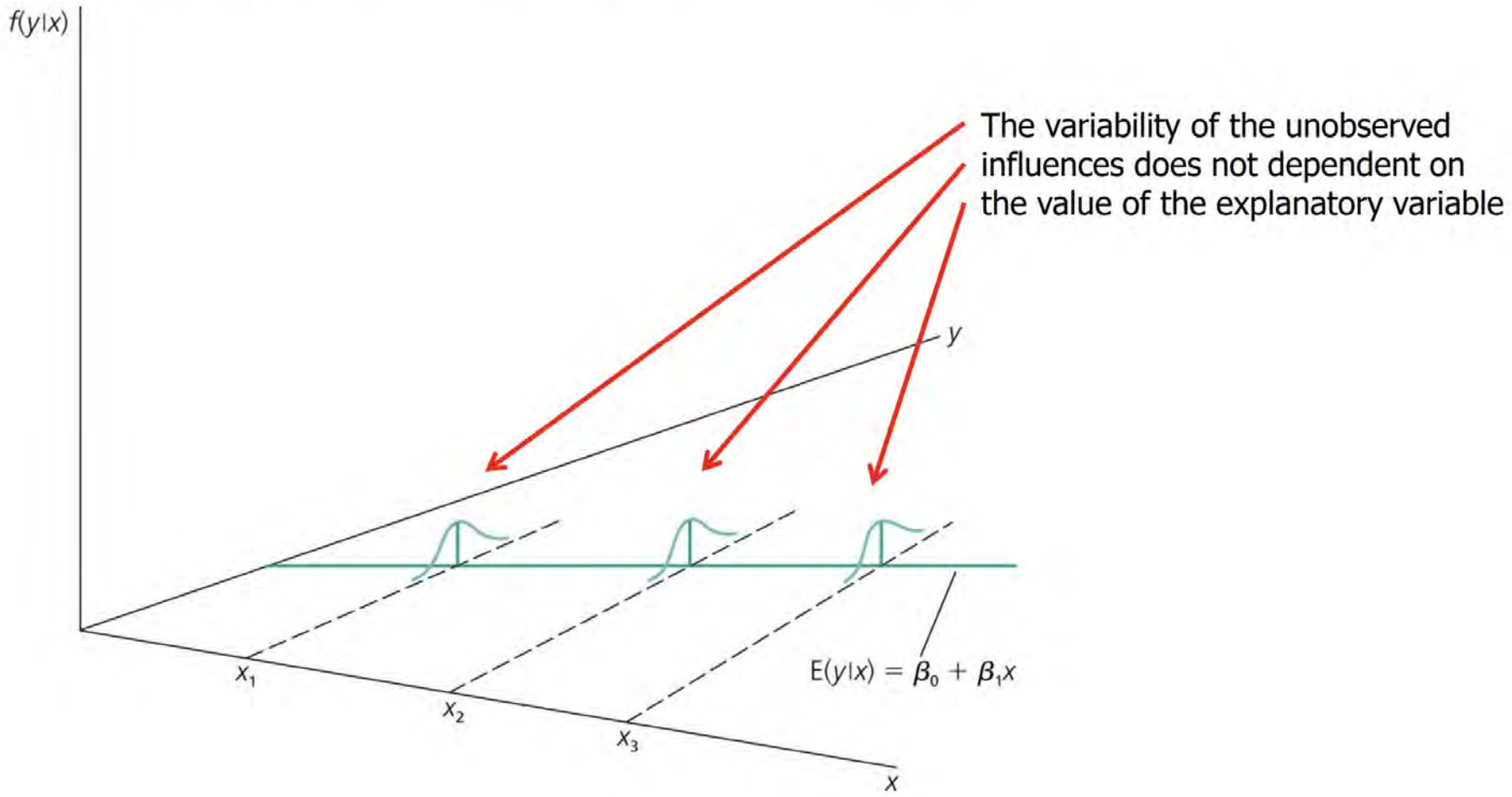
  with $\beta_1 > 0$. This means GPA of university study increases with high school score. If we impose SLR.5 then

$$Var(colGPA|hsGPA) = \sigma^2$$

  which means the variability in GPA of university study does not change with high school GPA.

Variances of the OLS estimators

- **Graphical illustration of homoskedasticity**



Variances of the OLS estimators

- **Example: Wage and Education**

$$E(wage|educ) = \beta_0 + \beta_1 educ$$

If we makes the homoscedasticity assumption, then

$$Var(wage|educ) = \sigma^2$$

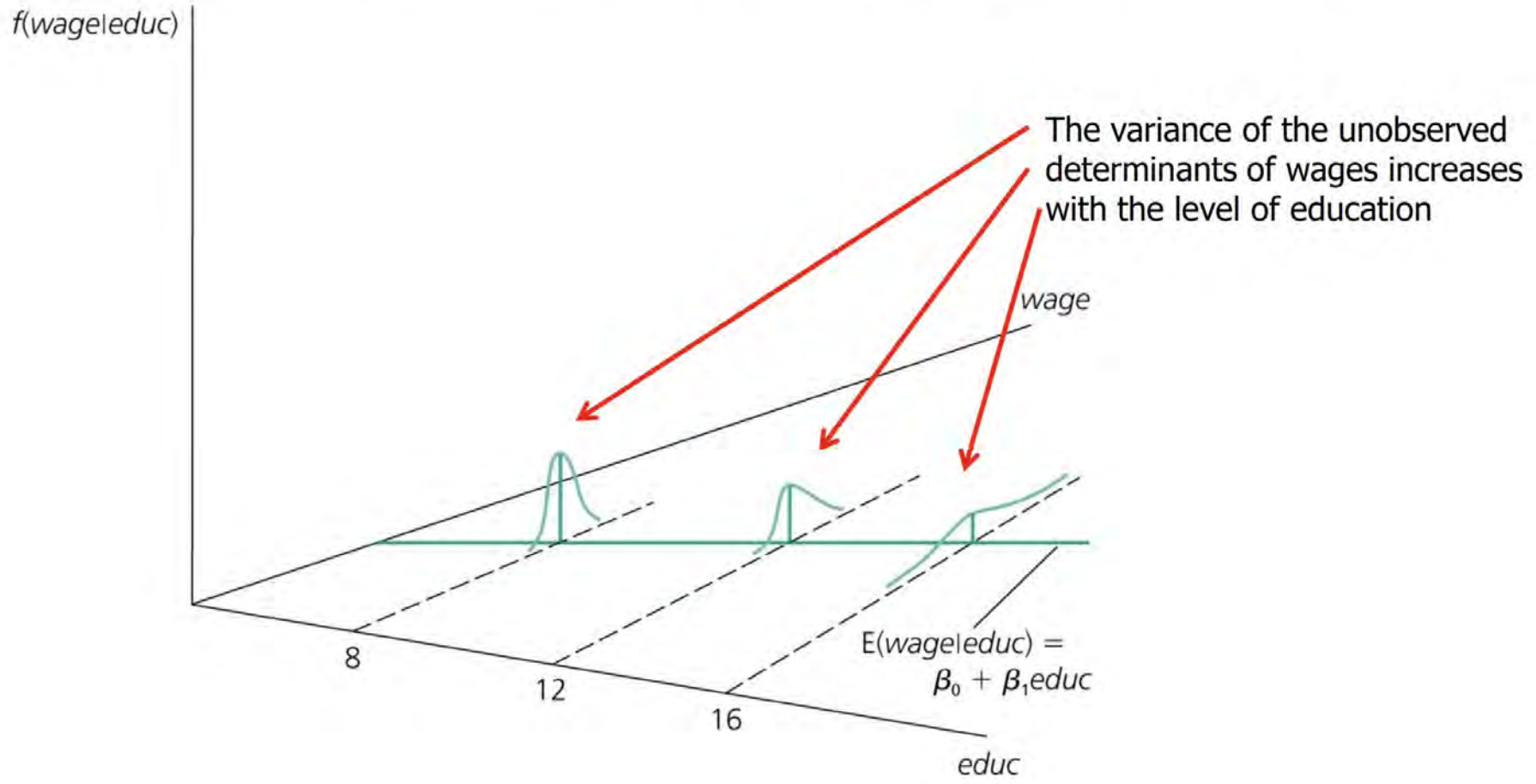
So, while average wage is allowed to increase with education level, the *variability* in wage is assumed to be constant across all education levels.

This is **not** realistic:

People with more education have a wider variety of interests and job opportunities, which could lead to more wage variability at higher levels of education

Variances of the OLS estimators

- **An example for heteroskedasticity: Wage and education**



Variances of the OLS estimators

- **Theorem 2.2 (Sampling Variances of OLS estimators)**

  Under assumptions SLR.1 – SLR.5:

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\sigma^2}{SST_x}$$

$$Var(\hat{\beta}_0) = \frac{\sigma^2 n^{-1} \sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\sigma^2 n^{-1} \sum_{i=1}^{n} x_i^2}{SST_x}$$

- **Conclusion:**

  The sampling variability of the estimated regression coefficients will be the higher the larger the variability of the unobserved factors, and the lower, the higher the variation in the explanatory variable

Variances of the OLS estimators

- To show this result, write, as before,

$$\hat{\beta}_1 = \beta_1 + \sum_{i=1}^{n} w_i u_i$$

where $w_i = (x_i - \bar{x})/SST_x$. We are treating the w_i as nonrandom in the derivation (conditional on $\{x_1, x_2, ..., x_n\}$).

$$\begin{aligned} Var(\hat{\beta}_1) &= Var\left(\sum_{i=1}^{n} w_i u_i\right) = \sum_{i=1}^{n} Var(w_i u_i) \\ &= \sum_{i=1}^{n} w_i^2 Var(u_i) = \sum_{i=1}^{n} w_i^2 \sigma^2 = \sigma^2 \sum_{i=1}^{n} w_i^2 \end{aligned}$$

since $Var(u_i) = \sigma^2$ for all i (SLR. 5)

Variances of the OLS estimators

- Now we have

$$\sum_{i=1}^{n} w_i^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{(SST_x)^2} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{(SST_x)^2}$$

$$= \frac{SST_x}{(SST_x)^2} = \frac{1}{SST_x}$$

so

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{SST_x}$$

- Conclusion:

1. As the error variance increases, that is, as σ^2 increases, so does $Var(\hat{\beta}_1)$. The more *noise* in the relationship between y and x (the larger variability in u), the harder it is to estimate β_1.

2. By contrast, more variation in $\{x_i\}$ is a good thing.

Variances of the OLS estimators

3. Notice that $SST_x/(n-1)$ is the sample variance in x. We can think of this as getting close to the population variance of x, σ_x^2, as n gets large. This means

$$SST_x \approx n\sigma_x^2$$

which means, as n grows, $Var(\hat{\beta}_1)$ shrinks at the rate $1/n$. This is why more data is a good thing: More data shrinks the sampling variance of our estimators.

- The standard deviation of $\hat{\beta}_1$ is the square root of the variance. So

$$sd(\hat{\beta}_1) = \frac{\sigma}{\sqrt{SST_x}}$$

Estimating the error variance

- **Estimating the error variance**

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{SST_x}$$

- We need a way to estimate σ^2. Recall that

$$\sigma^2 = E(u^2).$$

Therefore, if we **could** (actually can't) observe a sample on the **errors**, $\{u_i : i = 1, 2, ..., n\}$, an unbiased estimator of σ^2 would be

$$n^{-1} \sum_{i=1}^{n} u_i^2$$

Estimating the error variance

- **Errors vs. residuals**

  How about replacing each u_i with its "estimate", the OLS **residual** $\hat{u}_i$?

$$\begin{aligned} u_i &= y_i - \beta_0 - \beta_1 x_i \\ \hat{u}_i &= y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \end{aligned}$$

  then

$$\begin{aligned} \hat{u}_i &= y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i = (\beta_0 + \beta_1 x_i + u_i) - \hat{\beta}_0 - \hat{\beta}_1 x_i \\ &= u_i - (\hat{\beta}_0 - \beta_0) - (\hat{\beta}_1 - \beta_1) x_i \end{aligned}$$

  $E(\hat{\beta}_0) = \beta_0$ and $E(\hat{\beta}_1) = \beta_1$, but the estimators almost always differ from the population values in a sample.

Estimating the error variance

- What about this as an estimator of σ^2 ?

$$E(\hat{u}^2) = \frac{1}{n}\sum_{i=1}^{n}\hat{u}_i^2$$



Biased!

The estimator does not account for the two restrictions on the residuals, used to obtain $\hat{\beta}_0$ and $\hat{\beta}_1$:

$$n^{-1}\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$n^{-1}\sum_{i=1}^{n}x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Estimating the error variance

- The unbiased estimator of σ^2 uses a **degrees-of-freedom** adjustment.
  The residuals have only $n-2$ degrees-of-freedom, not n .

  The estimator used universally is

$$\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n}\hat{u}_i^2$$

- **THEOREM 2.3 (Unbiased Estimator of σ^2)**

  Under Assumptions SLR.1 to SLR.5

$$E(\hat{\sigma}^2) = \sigma^2$$

$$\hat{\sigma} = \sqrt{\hat{\sigma}^2} = \sqrt{SSR/(n-2)}$$

Estimating the error variance

- $\hat{\sigma}$ is called the **standard error of the regression**, which means it is an estimate of the standard deviation of the error in the regression.

Given $\hat{\sigma}$, we can now estimate the standard deviation of β_1 and β_0. The estimates of these are called the **standard errors** of the $\hat{\beta}$.

Plug in $\hat{\sigma}^2$ for the unknown σ^2

$$se(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{SST_x}} = \frac{\hat{\sigma}}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$$

It measures how precisely the regression coefficient is estimated.

Estimating the error variance

- **EXAMPLE: Return to Education**

  Data: WAGE2.DTA

$$\widehat{lwage} = \underset{(.008)}{0.083} + \underset{(.097)}{.584}\, educ$$

$$n = 526,\ R^2 = .186$$



The Simple Regression Model: Summary

- Theorem 2.1 (Unbiasedness of OLS)

$$SLR.1 - SLR.4 \quad \Rightarrow \quad E(\hat{\beta}_0) = \beta_0, \; E(\hat{\beta}_1) = \beta_1$$

- Interpretation of unbiasedness
  - The estimated coefficients may be smaller or larger, depending on the sample that is the result of a random draw
  - However, on average, they will be equal to the values that charac-terize the true relationship between y and x in the population
  - "On average" means if sampling was repeated, i.e. if drawing the random sample and doing the estimation was repeated many times
  - In a given sample, estimates may differ considerably from true values

The Simple Regression Model: Summary

- Theorem 2.2 (Variances of the OLS estimators)

Under assumptions SLR.1 – SLR.5:

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\sigma^2}{SST_x}$$

$$Var(\hat{\beta}_0) = \frac{\sigma^2 n^{-1} \sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\sigma^2 n^{-1} \sum_{i=1}^{n} x_i^2}{SST_x}$$

- Conclusion:
  - The sampling variability of the estimated regression coefficients will be the higher, the larger the variability of the unobserved factors, and the lower, the higher the variation in the explanatory variable

The Simple Regression Model: Summary

- Estimating the error variance

$$Var(u_i|x_i) = \sigma^2 = Var(u_i)$$

The variance of u does not depend on x, i.e. equal to the unconditional variance

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(\hat{u}_i - \bar{\hat{u}}_i)^2 = \frac{1}{n}\sum_{i=1}^{n}\hat{u}_i^2$$

One could estimate the variance of the errors by calculating the variance of the residuals in the sample; unfortunately this estimate would be biased

$$\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n}\hat{u}_i^2$$

An unbiased estimate of the error variance can be obtained by substracting the number of estimated regression coefficients from the number of observations

The Simple Regression Model: Summary

- Theorem 2.3 (Unbiasedness of the error variance)

$$SLR.1 - SLR.5 \quad \Rightarrow \quad E(\hat{\sigma}^2) = \sigma^2$$

- Calculation of standard errors for regression coefficients

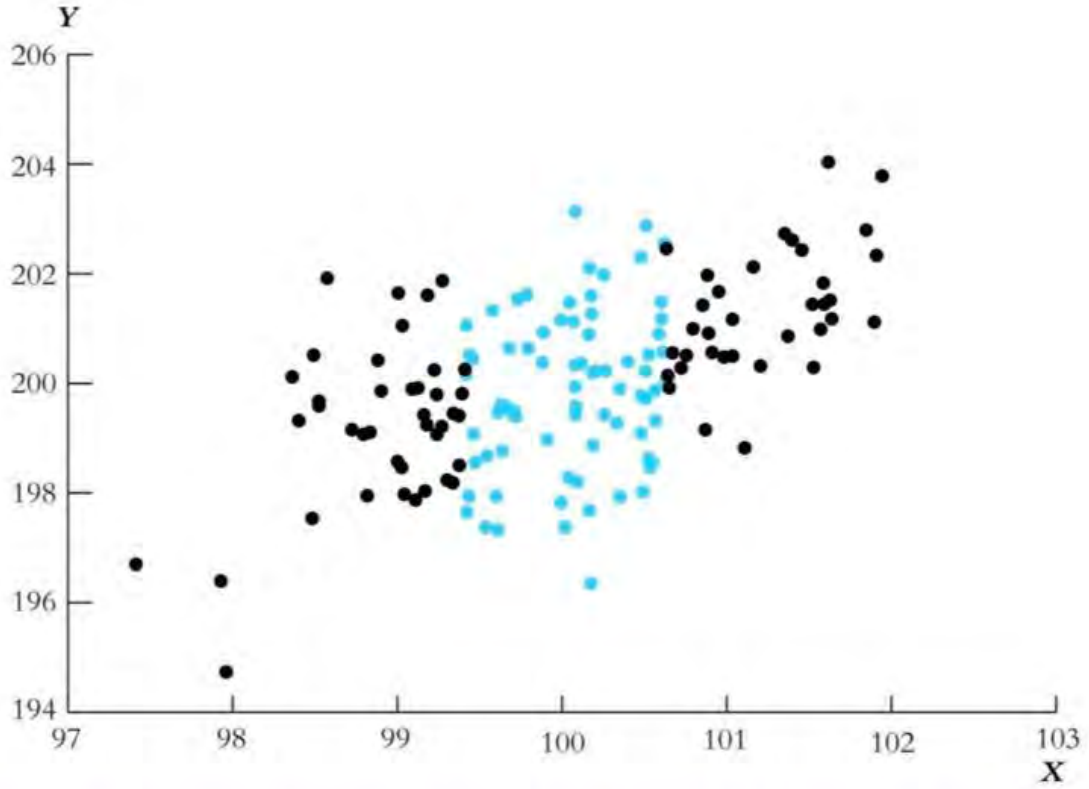
$$se(\hat{\beta}_1) = \sqrt{\widehat{Var}(\hat{\beta}_1)} = \sqrt{\hat{\sigma}^2 / SST_x}$$

Plug in $\hat{\sigma}^2$ for the unknown σ^2

$$se(\hat{\beta}_0) = \sqrt{\widehat{Var}(\hat{\beta}_0)} = \sqrt{\hat{\sigma}^2 n^{-1} \sum_{i=1}^{n} x_i^2 / SST_x}$$

The estimated standard deviations of the regression coefficients are called "standard errors." They measure how precisely the regression coefficients are estimated.

The Simple Regression Model: Summary

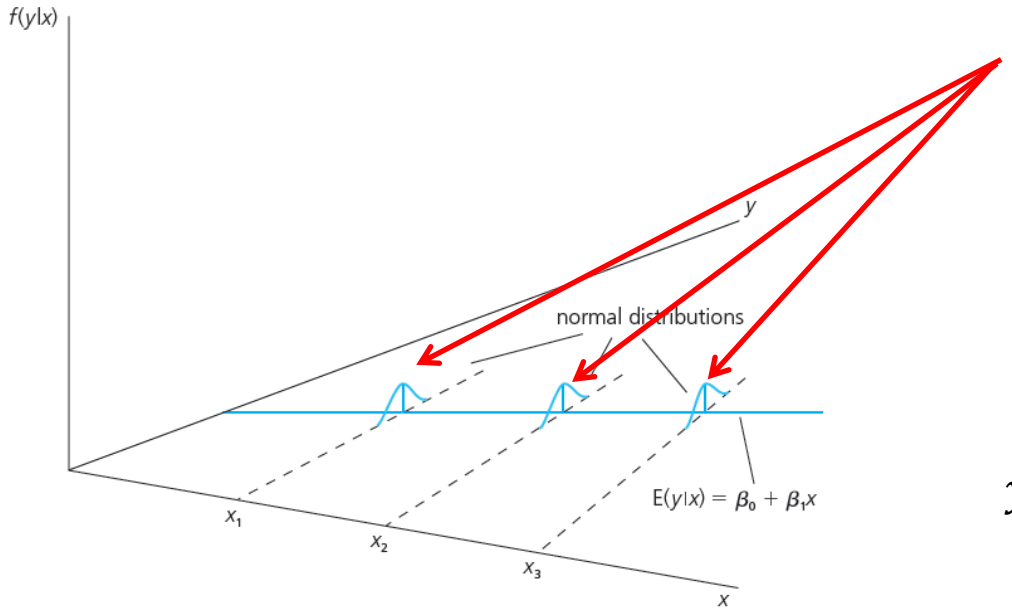


There are the same number of black and blue dots - using which would you get a more accurate regression line?

Testing

- Assumption SLR.6 (Normality of error terms)

$u_i \sim \text{Normal}(0, \sigma^2)$ independently of x_i



It is assumed that the unobserved factors are normally distributed around the population regression function.

The form and the variance of the distribution does not depend on any of the explanatory variables.

It follows that:

$$y|x \sim Normal(\beta_0 + \beta_1 x, \sigma^2)$$

Testing

- Theorem (Normal sampling distributions)

Under assumptions SLR.1 – SLR.6:

$$\hat{\beta}_j \sim \text{Normal}(\beta_j, Var(\hat{\beta}_j))$$

The estimators are normally distributed around the true parameters with the variance that was derived earlier

$$\frac{\hat{\beta}_j - \beta_j}{sd(\hat{\beta}_j)} \sim \text{Normal}(0, 1)$$

The standardized estimators follow a standard normal distribution

Testing

- Theorem (t-distribution for the standardized estimators)

Under assumptions SLR.1 – SLR.6:

$$\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \sim t_{n-k-1}$$

If the standardization is done using the estimated standard deviation (= standard error), the normal distribution is replaced by a t-distribution

Note: The t-distribution is close to the standard normal distribution if n-k-1 is large.

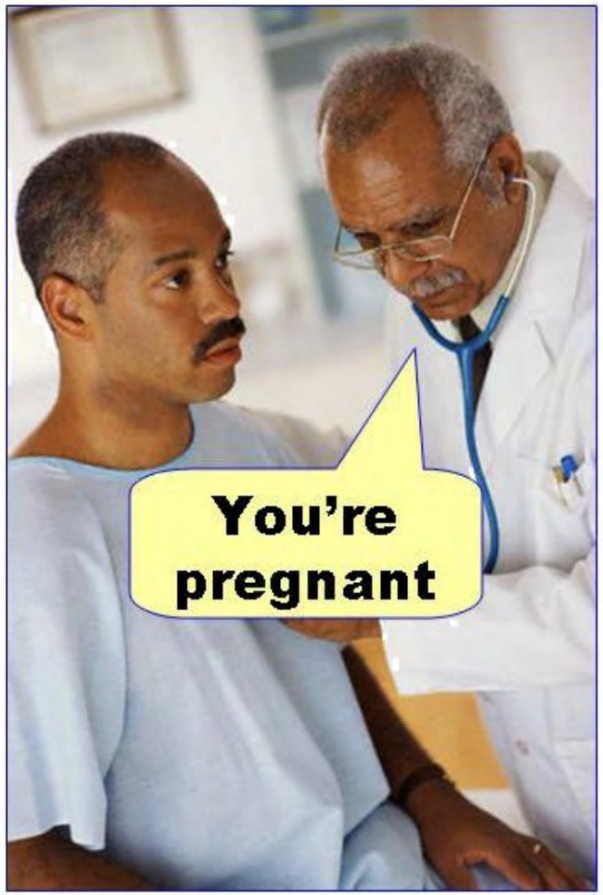
- Null hypothesis (more general hypotheses will be discussed later)

$$H_0: \ \beta_j = 0$$

The population parameter is equal to zero, i.e. after controlling for the other independent variables, there is no effect of x_j on y

Testing

Type I error
(false positive)



Type II error
(false negative)



Testing

- t-statistic (or t-ratio)

$$t_{\hat{\beta}_j} \equiv \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}$$

The t-statistic will be used to test the above null hypothesis. The farther the estimated coefficient is away from zero, the less likely it is that the null hypothesis holds true. But what does "far" away from zero mean?

This depends on the variability of the estimated coefficient, i.e. its standard deviation. The t-statistic measures how many estimated standard deviations the estimated coefficient is away from zero.

- Distribution of the t-statistic if the null hypothesis is true

$$t_{\hat{\beta}_j} \equiv \hat{\beta}_j / se(\hat{\beta}_j) = (\hat{\beta}_j - \beta_j)/se(\hat{\beta}_j) \sim t_{n-k-1}$$

- Goal: Define a rejection rule so that, if it is true, H0 is rejected only with a small probability (= significance level, e.g. 5%)

Testing

t_{n-2} v.s. $N(0,1)$

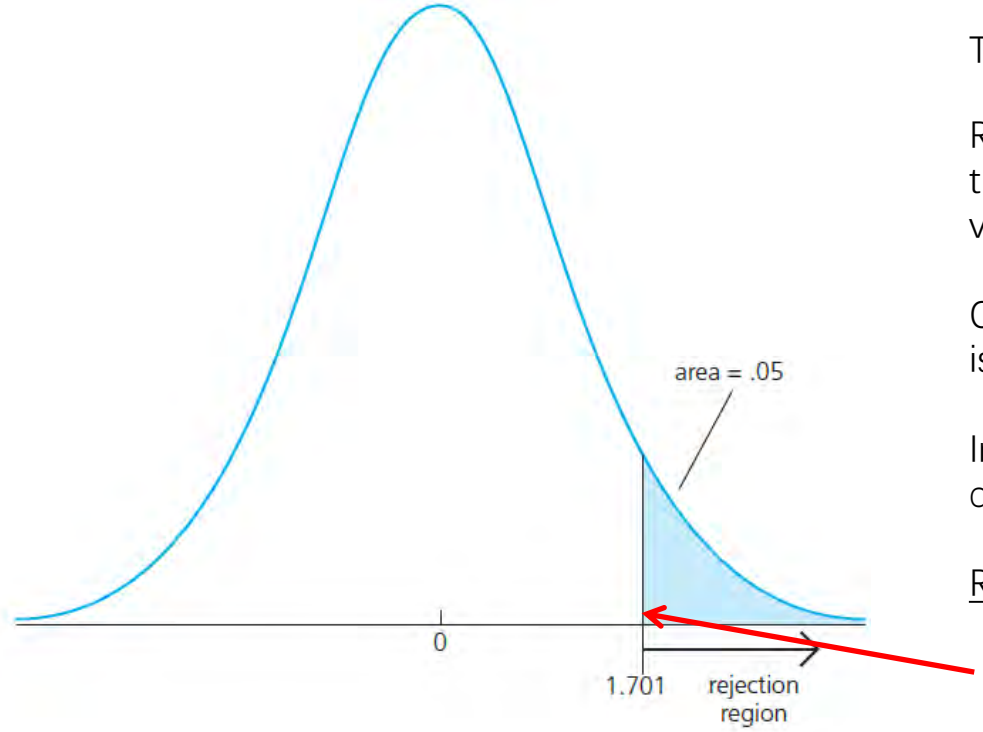
- In fact, the difference between the t_{n-2} and N(0,1) critical values is negligible when n is large

Degrees of Freedom (n-2)	10	20	30	60	∞
5% t Distribution Critical Value	2.23	2.09	2.04	2.00	1.96

- Statistical software typically uses the t distribution to compute p values - but this is irrelevant when the sample size is moderate or large ($n \geq 50$)
- You can challenge the result when n is too small, e.g. $n = 20$. In this course, we will focus on the large$-n$ approximation

Testing

- Testing against one-sided alternatives (greater than zero)



Test $H_0 : \beta_j = 0$ against $H_1 : \beta_j > 0$

Reject the null hypothesis in favour of the alternative hypothesis if the estimated coefficient is "too large" (i.e. larger than a critical value).

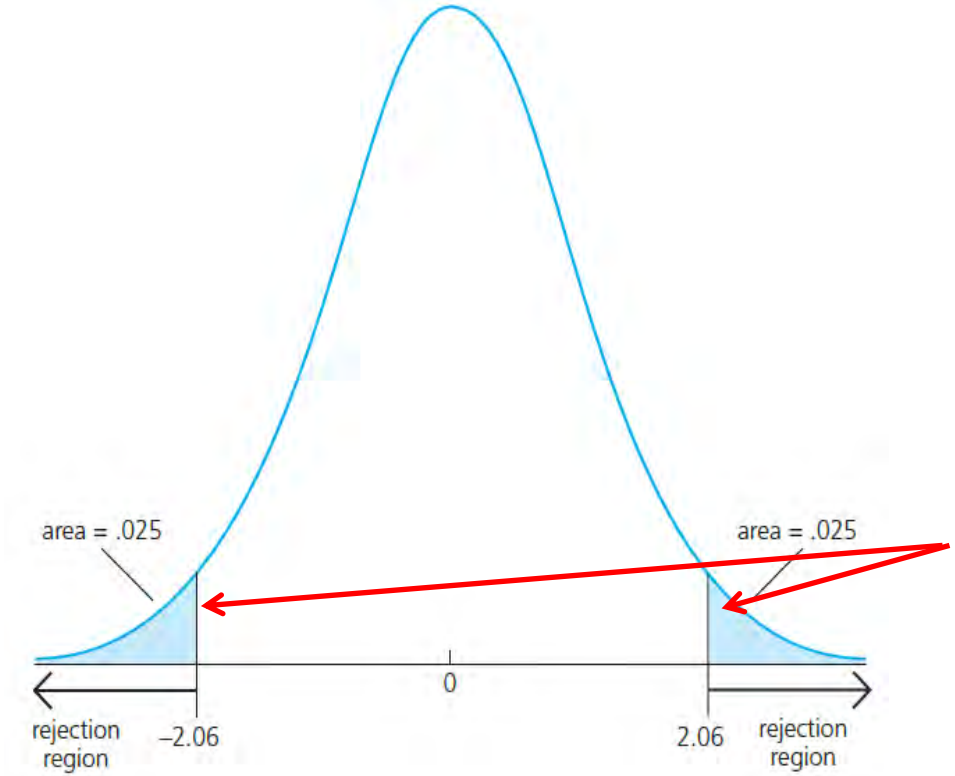
Construct the critical value so that, if the null hypothesis is true, it is rejected in,for example, 5% of the cases.

In the given example, this is the point of the t-distribution with 28 degrees of freedom that is exceeded in 5% of the cases.

Reject if t-statistic is greater than 1.701

Testing

- Testing against two-sided alternatives



Test $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$

Reject the null hypothesis in favour of the alternative hypothesis if the absolute value of the estimated coefficient is too large.

Construct the critical value so that, if the null hypothesis is true, it is rejected in, for example, 5% of the cases.

In the given example, these are the points of the t-distribution so that 5% of the cases lie in the two tails.

Reject if absolute value of t-statistic is greater than 2.06

Properties of OLS on any sample of data

- Goodness-of-Fit

"How well does the explanatory variable explain the dependent variable?"

$$y_i = \hat{y}_i + \hat{u}_i$$

- Measures of Variation

$$SST \equiv \sum_{i=1}^{n} (y_i - \bar{y})^2$$

Total sum of squares, represents total variation in the dependent variable

$$SSE \equiv \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

Explained sum of squares, represents variation explained by regression

$$SSR \equiv \sum_{i=1}^{n} \hat{u}_i^2$$

Residual sum of squares, represents variation not explained by regression

Properties of OLS on any sample of data

- **Decomposition of total variation**

$$\begin{aligned} SST &= \sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}[(y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})]^2 \\ &= \sum_{i=1}^{n}[\hat{u}_i + (\hat{y}_i - \bar{y})]^2 \end{aligned}$$

using the fitted values and residuals are uncorrelated, can show:

$$SST = SSE + SSR$$

Total variation — Explained part — Unexplained part

Properties of OLS on any sample of data



- **Goodness-of-fit measure (R-squared)**

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$$

R-squared measures the fraction of the total variation that is explained by the regression

$$0 \leq R^2 \leq 1$$

- $R^2 = 0$ means no linear relationship between y_i and x_i. $R^2 = 1$ means a perfect linear relationship
- As R^2 increases, the y_i are closer and closer to falling on the OLS regression line
- R^2 is a useful summary measure but tells us nothing about causality. Having a high R^2 is neither necessary nor sufficient to infer causality.

Properties of OLS on any sample of data

- CEO Salary and return on equity

$$\widehat{salary} = 963.191 + 18.501\ roe$$

$$n = 209, \quad R^2 = 0.0132$$

The regression explains only 1.3% of the total variation in salaries

- Voting outcomes and campaign expenditures

$$\widehat{voteA} = 26.81 + 0.464\ shareA$$

$$n = 173, \quad R^2 = 0.856$$

The regression explains 85.6% of the total variation in election outcomes

- Caution: A high R-squared does not necessarily mean that the regression has a causal interpretation!

Units of Measurement and Functional Form

- **Units of Measurement**

  It is very important to know how y and x are measured in order to interpret regression functions.

  Example: CEO Salary and return on equity

  $$\widehat{salary} = 963.191 + 18.501\ roe$$

  Salary in thousands of dollars

  Return on equity (In percent) of the CEO's firm

- What if we measure roe as a decimal, rather than a percent? Define

  $$roedec = roe/100$$

  Does it change the regression results?

Units of Measurement and Functional Form

- **The original regression**

$$\widehat{salary} = 963.191 + 18.501\, roe$$
$$n = 209, \quad R^2 = 0.0132$$

- **The new regression**

$$\widehat{salary} = 963.191 + 1,850.1\, roedec$$
$$n = 209, \quad R^2 = 0.0132$$

Now one percentage point change in roe is the same as 0.01 percentage points change in $roedec$ so we get the same effect as before.

Units of Measurement and Functional Form

- **Changing units of measurement for dependent variable**

  What if we measure salary in dollars, rather than thousands of dollars

$$salarydol = 1,000 \cdot salary$$

The new regression:

$$\begin{aligned} \widehat{salarydol} &= 963,191 + 18,501\ roe \\ n &= 209,\ R^2 = .0132 \end{aligned}$$

Changing units of measurement for both dependent and independent variable?

Units of Measurement and Functional Form

- **Using the Natural Logarithm in Simple Regression**

  Recall the wage example:

$$\widehat{wage} = -0.90 + 0.54\ educ$$
$$n = 526,\ R^2 = 0.1648$$

- Problem of this linear model: the dollar value of another year of schooling is constant. We expect additional years of schooling to be worth more, in dollar terms, than previous years.

- **How can we incorporate an increasing effect?**

  One way is to postulation a constant **percentage** effect. We can approximate percentage changes using the natural log.

Units of Measurement and Functional Form

- **Regression of log wages on years of eduction**

$$\log(wage) = \beta_0 + \beta_1 educ + u$$

Natural logarithm of wage

- This changes the interpretation of the regression coefficient:
  Holding u fixed,

$$\Delta \log(wage) = \beta_1 \Delta educ$$

Useful result from calculus

$$100 \cdot \Delta \log(wage) \approx \%\Delta wage$$

Percentage change of wage

therefore

$$\beta_1 = \frac{\Delta \log(wage)}{\Delta educ} \approx \frac{100^{-1} \cdot \%\Delta wage}{\Delta educ}$$

Units of Measurement and Functional Form

- **Using the Natural Logarithm in Simple Regression (cont.)**

$$100\beta_1 \approx \%\Delta wage \text{ when } \Delta educ = 1$$

In this example, $100\beta_1$ is often called the return to education. This measure is free of units of measurements.

- **The OLS regression**

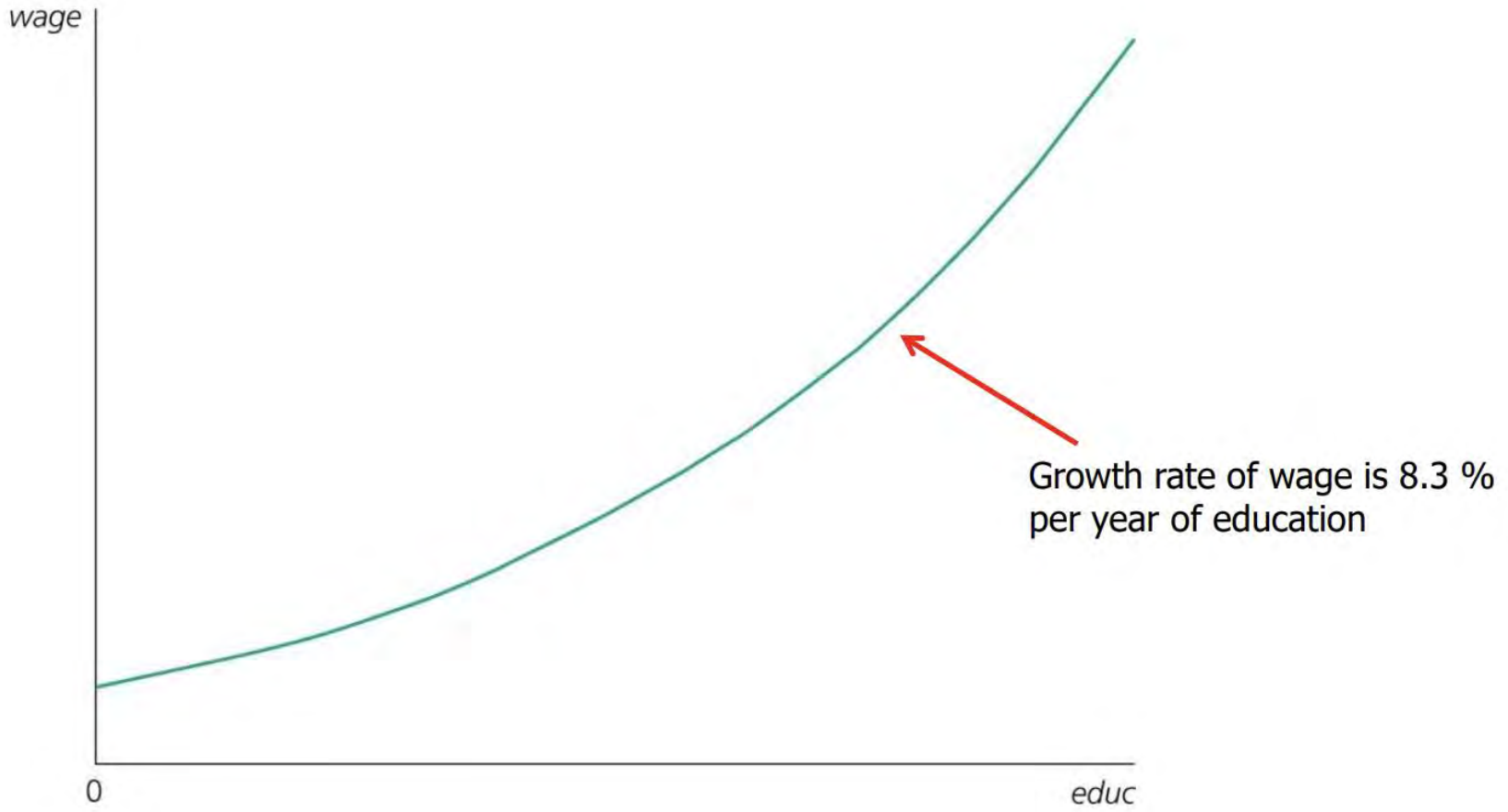
$$\widehat{\log}(wage) = 0.584 + 0.083\ educ$$

$$n = 526, \quad R^2 = 0.1858$$

The wage increases by 8.3 % for every additional year of education (= return to education)

Units of Measurement and Functional Form

- **The relationship between wage and education**



Units of Measurement and Functional Form

- **Constant elasticity model**

  For example, CEO salary and firm sales.

$$\log(salary) = \beta_0 + \beta_1 \log(sales) + u$$

  Then

$$\beta_1 \approx \frac{\%\Delta salary}{\%\Delta sales}$$

- **The OLS regression**

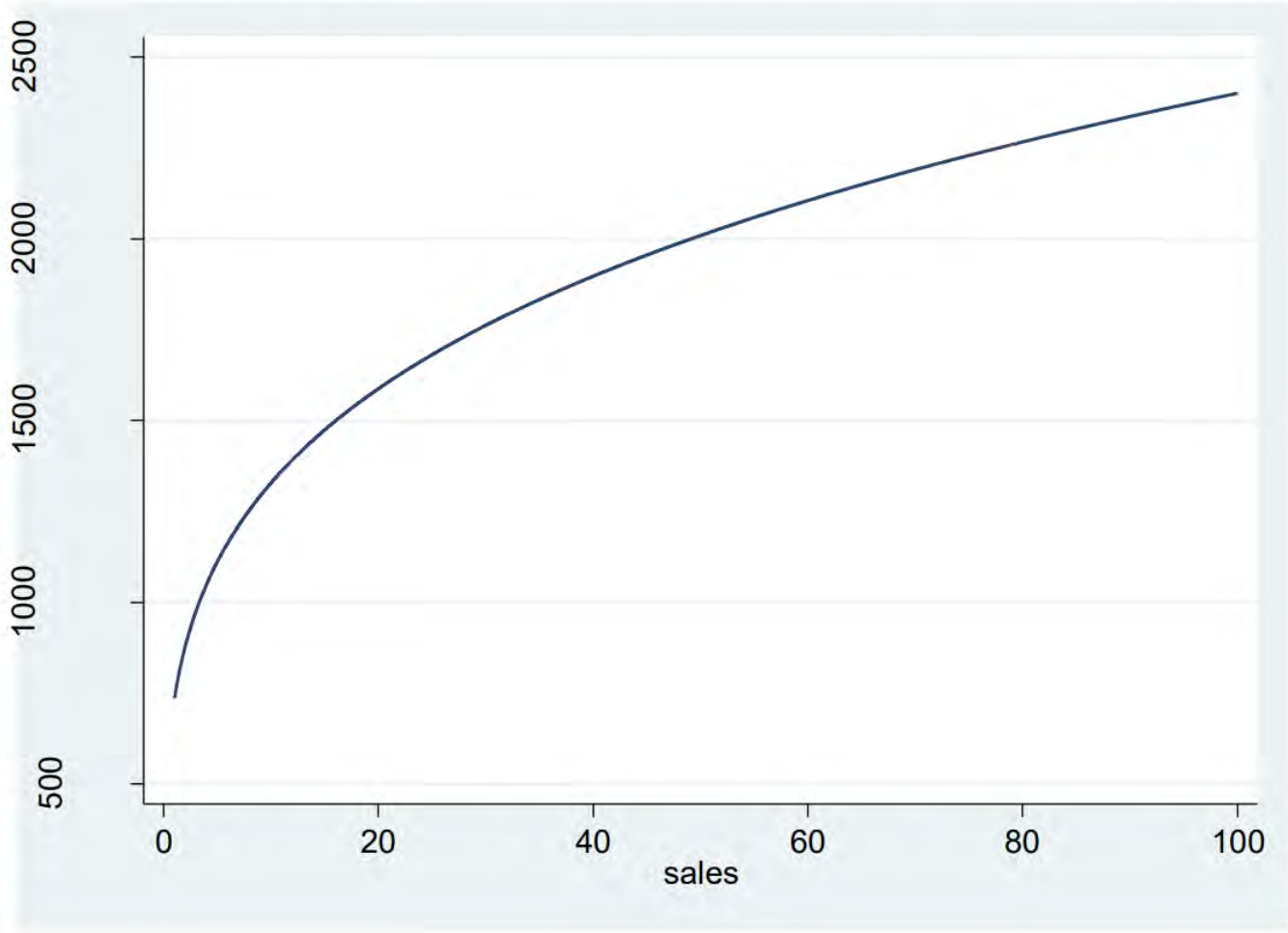
  Data CEOSAL1.dta

$$\widehat{\log}(salary) = 4.822 + 0.257 \log(sales)$$

The salary increases by 0.257 % for every 1% increase in sales

Units of Measurement and Functional Form

- **The relationship between CEO salary and firm sales.**



Units of Measurement and Functional Form

- **Summay of Functional Forms involving Logarithms**

Model	Dep. Var.	Indep. Var.	β_1
Level-Level	y	x	$\Delta y = \beta_1 \Delta x$
Level-Log	y	$\log(x)$	$\Delta y = (\beta_1/100)\%\Delta x$
Log-Level	$\log(y)$	x	$\%\Delta y = (100\beta_1)\Delta x$
Log-Log	$\log(y)$	$\log(x)$	$\%\Delta y = \beta_1 \%\Delta x$